



Generalizable Dynamics Models for Deformable Objects: Tool-Agnostic Model for Tool Geometry Design

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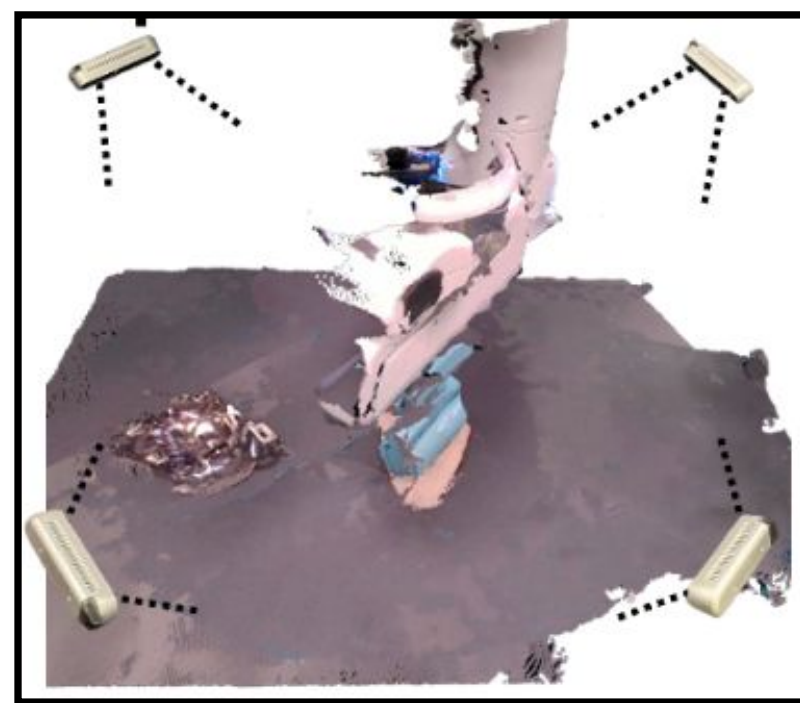


Project Page

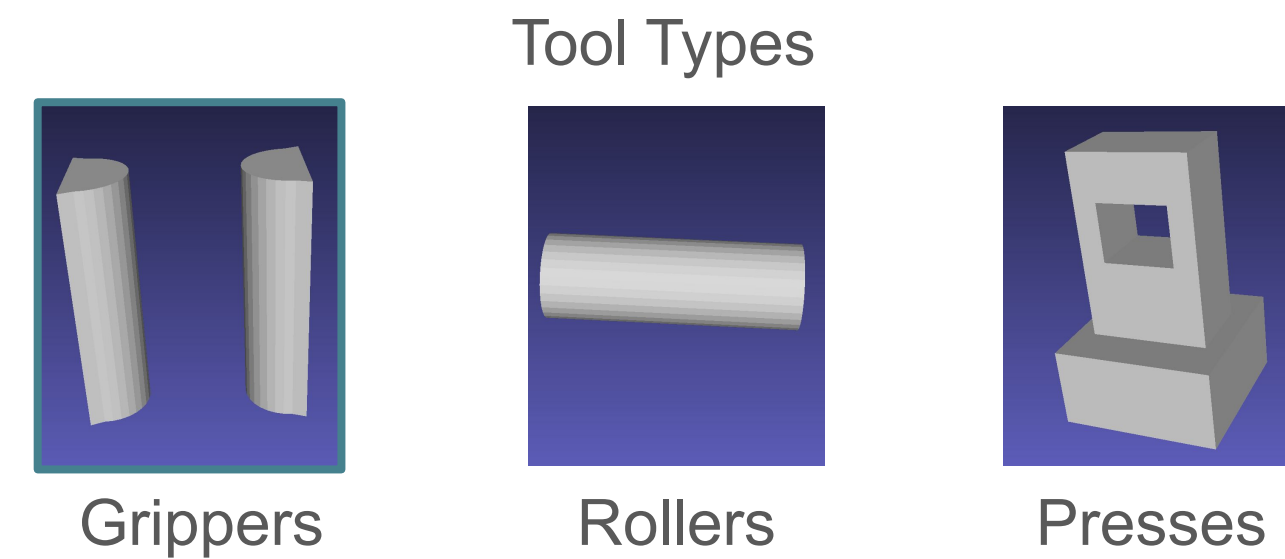
Motivation

- Model-based planning needs dynamics that stay **accurate across many scene configurations**.
- For elastoplastic objects, **analytic models** are too inaccurate.
- Prior tool-design methods have **limited shape expressivity**, relying on shape templates or cage-based deformations.

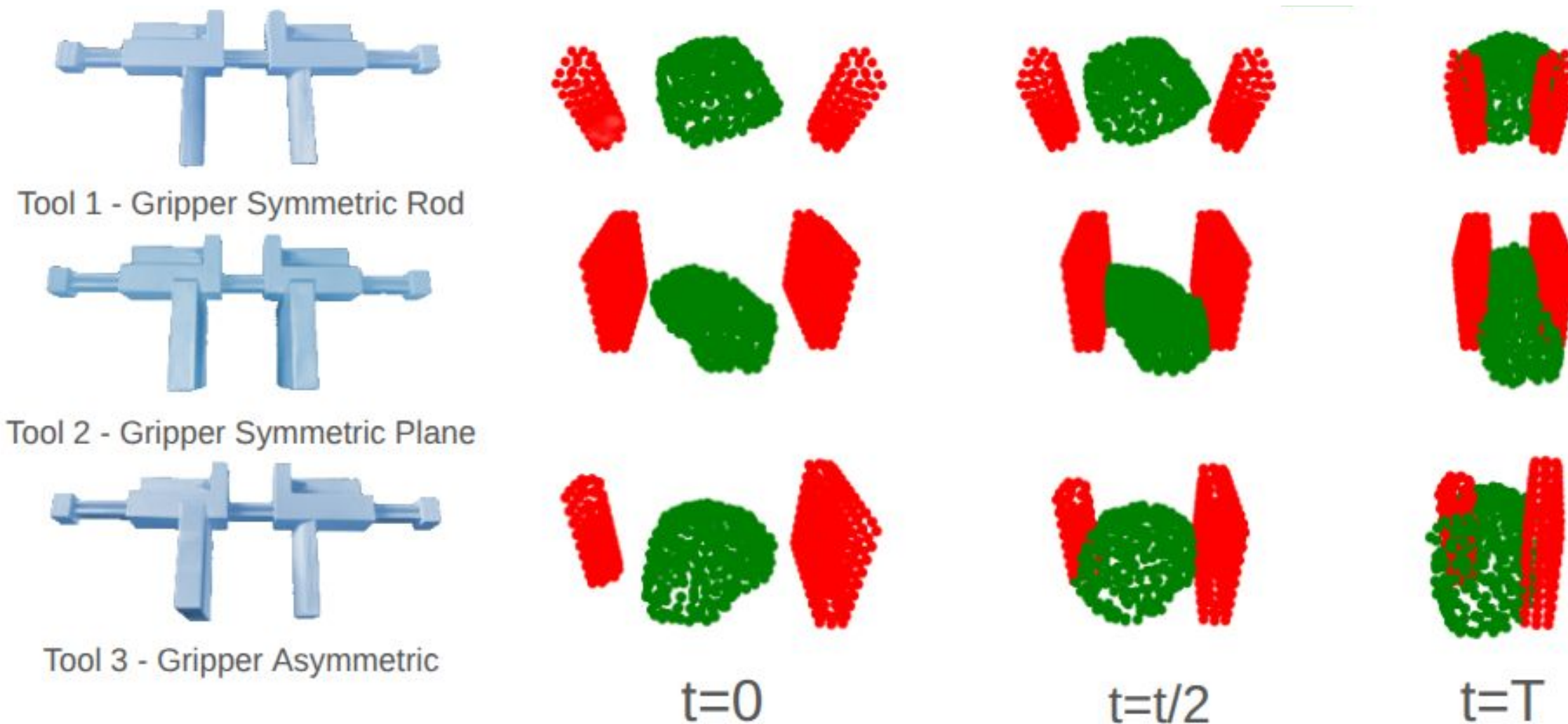
Model Training



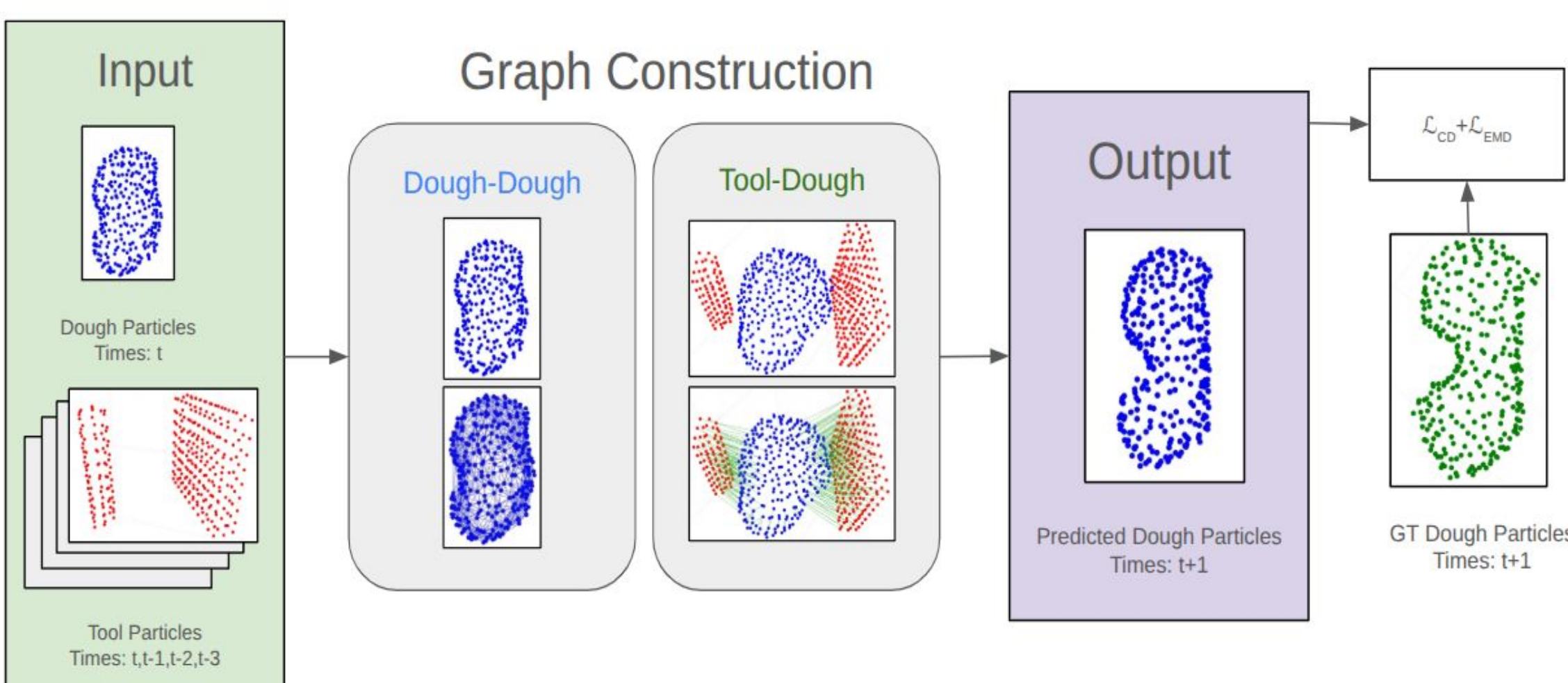
Input Data : Dough and tool point cloud sequence from multi-view RGBD.



Sampled trajectories collected across different gripper geometries (rows).



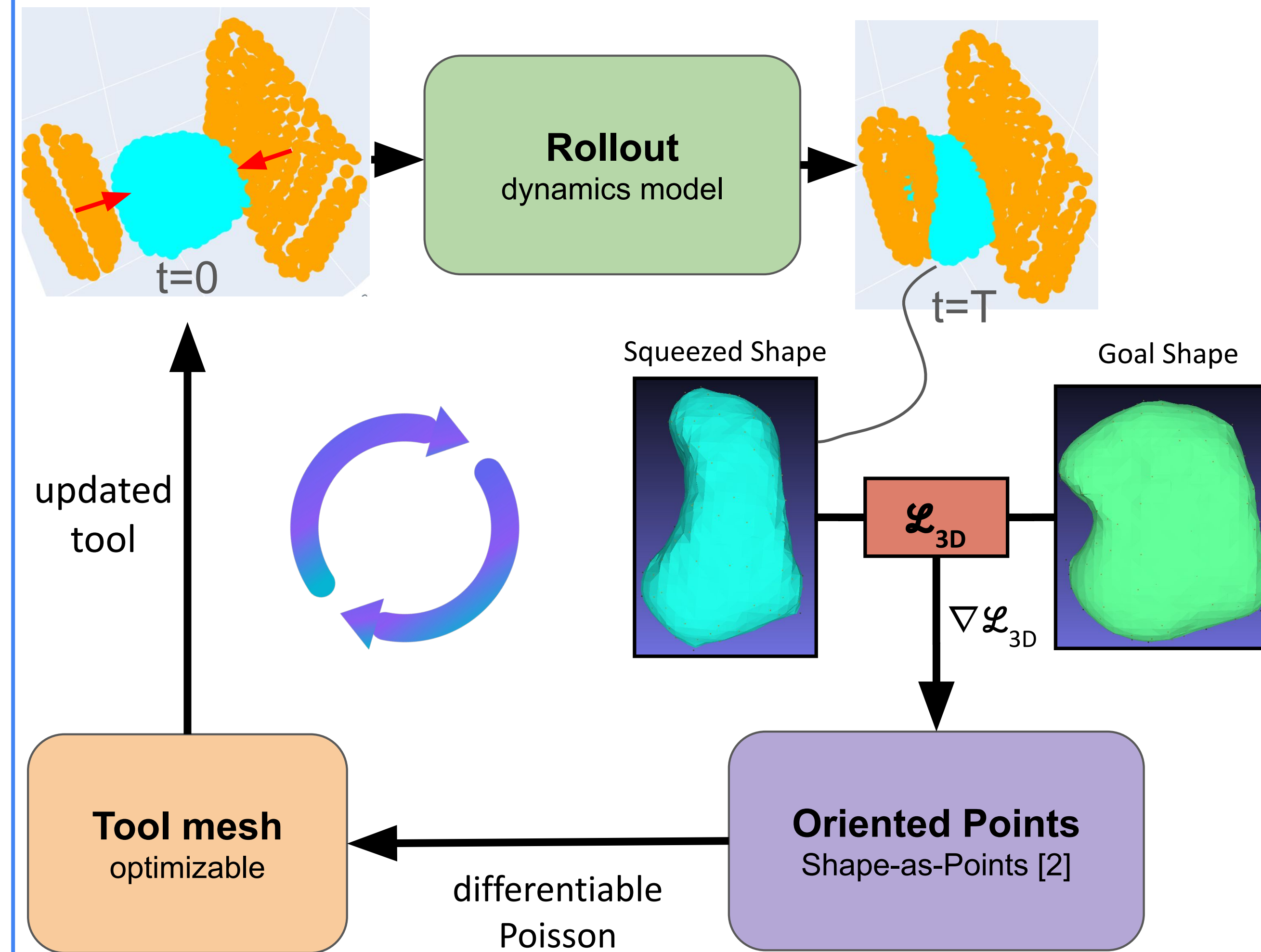
GNN Dynamics Model



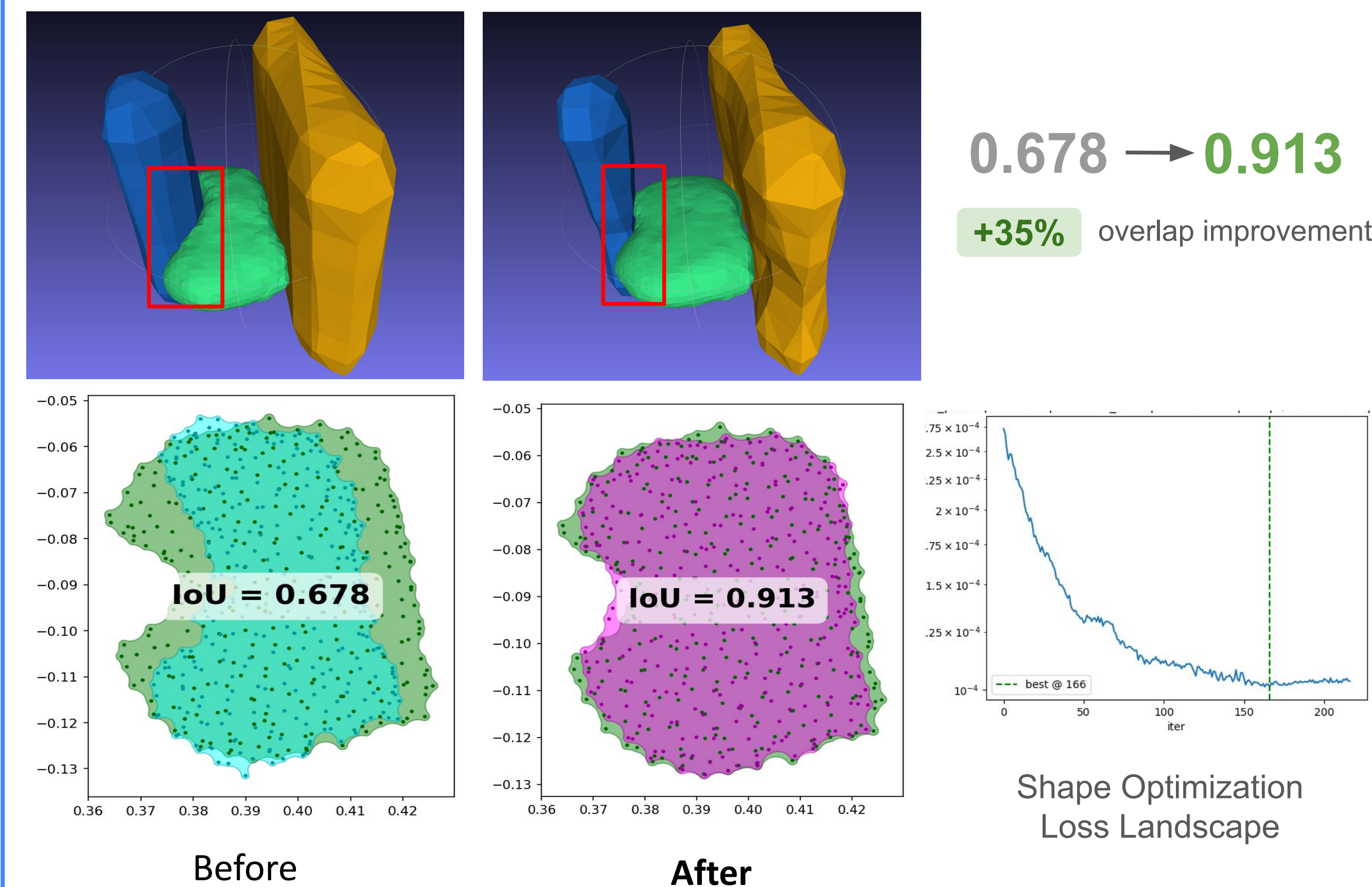
- Per-particle tool SDF feature for surface information.
- Tool connectivity radius jitter augmentation.
- Particle tool penetration augmentation using SDF gradients.

Tool Shape Optimization

Optimize tool shape iteratively to match dough goal shape!

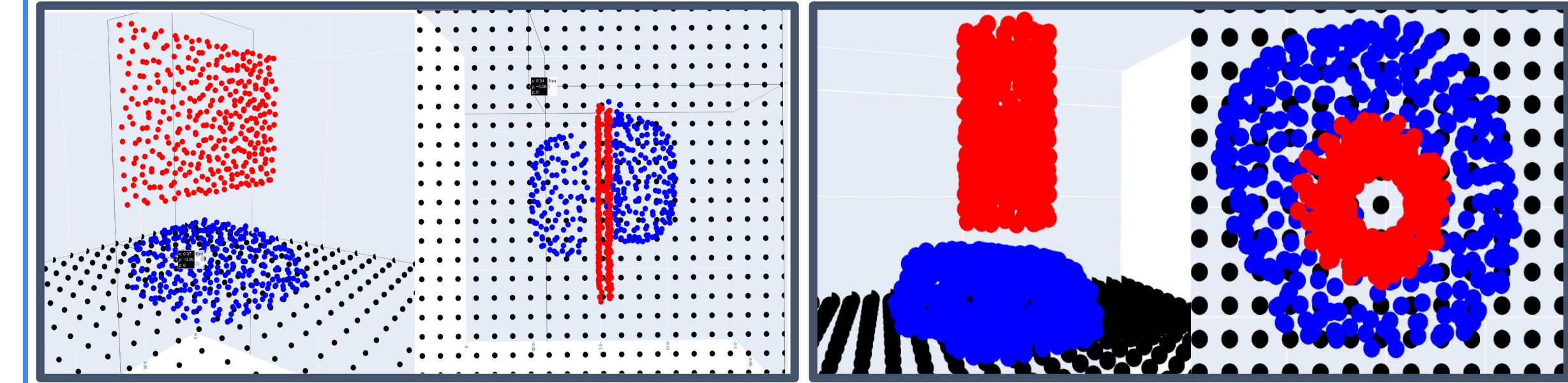


Shape Optimization Results



Dynamics Model on OOD Shape

Trained on grippers (pinching), model is tested on these unseen tools.



Multi-tool Training Improves In-distribution

# Train tools	CD↓	EMD↓	HD↓	SDF↓
1 gripper [1]	3.44	2.42	9.71	0.99
grippers	3.56	2.33	9.60	0.53
grippers + rollers	3.49	2.23	9.40	0.91
grippers + rollers + presses	3.24	2.22	9.31	0.55

Metrics for tool 3 (in mm). Metrics: Chamfer (CD), Earth Mover's (EMD), Hausdorff (HD) distance, and signed-distance penetration (SDF).

OOD Generalization on Held-out Shape

# Train tools	CD↓	EMD↓	HD↓	SDF↓
1 gripper	3.44	2.42	9.71	0.98
All tool types [1]	3.86	2.75	11.95	1.87
All tool types (ours)	3.67	2.67	11.07	0.31

Inflate-and-Recover Experiment

Inflation %	Inflated / GT (CD %)	Optimized / GT (CD %)
25	110.9	101.2
50	122.0	101.2
100	145.9	103.5

Conclusion

- Multi-tool training leads to generalization to unseen tool shapes.
- Improved reconstruction and fewer penetrations on real data.
- A differentiable dynamics model for high-dim tool design.

References

- Shi, H., Xu, H., Clarke, S., Li, Y. and Wu, J., 2023. Robocook: Long-horizon elasto-plastic object manipulation with diverse tools, CoRL 2023.
- Peng, S., Jiang, C., Liao, Y., Niemeyer, M., Pollefeys, M. and Geiger, A., 2021. Shape as points: A differentiable poisson solver, NeurIPS 2021.